



Building Image-Processing Embedded Systems Using Python

Part—3

The third and final part of this series shows you how to use the Pyserial module, Arduino, and finally, how to combine everything learnt so far to build a full-fledged embedded system for image processing. In the previous article, I had covered the use of the OpenCV library for image-processing via the Python interface.

As I had mentioned in the first article of this series, Pyserial is a library that provides Python support for serial communication. Now, let's explore how to use it and send data to other devices via serial communication. First fire up the Python interpreter (run *python* in a terminal). Then import the Pyserial module (*import serial*). If this gives no errors, we're ready—else you may need to install the package with the module. Now, since laptops don't have serial ports, I used a USB-to-serial converter (db9) for this demo. I shorted the Data Transmitting (pin 3) and Data Receiving (pin 2) pins, so whatever information I would send via the *pyserial* module, I would get back. Given below is the code (*Pyserial.py*) to illustrate this. You can download the

source (for all code in this article) from http://linuxforu.com/article_source_code/mar12/embedded_systems.zip.

```
#usr/bin/python
import serial
ser=serial.Serial('/dev/ttyUSB1',19200)
ser.write("Jayneil Dalal\n")
a=ser.readline()
print(a)
```

The first line opens the desired serial port and sets the baud rate. The second line writes the data to the serial port; the third reads the data from the serial port and stores it in a